DR. ABHIK ROY

EDUCATION

Western Michigan University
Ph.D. in Program Evaluation — Kalamazoo, MI

Dissertation. Building an Evaluation Model of Academic Advising's Impact on Progression, Persistence, and Retention Within University Settings

Michigan Technological University
M.S. in Mathematics — Houghton, MI

Thesis. Quotient Rings of the Eisenstein Integers

West Virginia Wesleyan College
B.S. in Mathematics — Buckhannon, WV

Report. 4-Cell Embedding on a n-genus Torus

PROFESSIONAL EXPERIENCE

Independent Evaluation Consultant
Present | 2023

Bloomington, IN

- Create static (fixed), active (automated or animated), and interactive (user-driven) data visualizations in R and Shiny to communicate findings across multiple audiences.
- Develop and validate survey instruments, implement mixed-methods studies, and author federal reporting documents.
- Lead evaluation studies, conduct data analysis, and report findings for federally funded programs in health and education.
- Partner with program staff and academic researchers to design and carry out evaluations that inform practice and policy.
- Prepare journal manuscripts based on evaluation findings for submission to peer-reviewed publications.
- Support research and evaluation for federally, state, and foundation-funded initiatives in health, education, and community development, including HRSA's Healthy Start Initiative, an NSF-supported teacher preparation program, and other multi-site applied studies.

Assistant Professor
2023 | 2016

West Virginia University — Morgantown, WV

- Authored ten peer-reviewed articles and spoke at more than twenty national conferences.
- Contributed to a dozen program evaluations, leading several and supporting analysis on others; most were funded by federal grants in education or health.
- Taught courses in evaluation, research methods, measurement, and survey design to several hundred students, with an emphasis on applied data science.
- Supported three master's candidates and one doctoral student from proposal to defense, providing ongoing support with research design, coding, and writing.
- Served on committees at the graduate (master's and doctoral), program, and college levels.; introduced new courses in data visualization and social network analysis.

CONTACT INFORMATION

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EXPERTISE

Program evaluation integrating traditional and machine learning methods

Quantitative, qualitative, and mixed-methods research

R programming

Statistical modeling and inference

Data visualization

Survey methodology spanning design, deployment, and analysis

Interpretive and computational text analysis

Web application development and reporting using blogdown, rmarkdown, Shiny

Social network analysis

Technical writing and markup using $\LaTeX$

TECHNICAL SKILLS

Methods: Longitudinal analysis, survival analysis, predictive modeling, NLP, topic modeling, clustering

Tools: Qualtrics, Shiny, Tableau, SQL

2016 | 2014 — Data Analyst
University of Kansas — Lawrence, KS

- Merged surveys, interviews, and registrar data to study how advising affected persistence, retention, and graduation.
- Ran needs assessments and participatory evaluations for advising, student-life, and support units.
- Distributed questionnaires to 300 students and staff; achieved response rates as high as 91%.
- Built predictive and survival models that flagged at-risk students, supporting tactics that raised retention by as much as five percentage points.
- Synthesized findings from Delphi panels, interviews, survey metrics, and time-to-event analysis into a unified framework for evaluating advising quality.
- Automated data pulls from the Oracle warehouse and refreshed Tableau dashboards with new R scripts.

PUBLICATIONS

2024 — Dey, K., Rahman, M. T., **Roy, A.**, Pyrialakou, V. D., Martinelli, D., Fraustino, J. D., Deskins, J., Rambo-Hernandez, K. E., & Plein, L. C. (2024). Experiences and perceptions of engineering students towards a cross-disciplinary course using sentiment analysis. *Journal of Civil Engineering Education, 150*(3). https://doi.org/10.1061/jceecd.eieng-1976

2023 — Kale, U., Kooken, A., Yuan, J., & **Roy, A.** (2023). Teaching science via computational thinking? Enabling future science teachers' access to computational thinking. *Contemporary Issues in Technology and Teacher Education, 23*(3). https://citejournal.org/volume-23/issue-3-23/science/teaching-science-via-computational-thinking-enabling-future-science-teachers-access-to-computational-thinking

Curtis, R., **Roy, A.**, Lewis, N., Dooty, E. N., & Mikalik, T. (2023). Program evaluation standards for utility facilitate stakeholder internalization of evaluative thinking in the West Virginia Clinical Translational Science Institute. *Journal of MultiDisciplinary Evaluation, 19*(43), 49–65. https://doi.org/10.56645/jmde.v19i43.831

2022 — Kale, U., Yuan, J., & **Roy, A.** (2022). Thinking processes in Code.org: A relational analysis approach to computational thinking. *Computer Science Education, 33*(4), 545–566. https://doi.org/10.1080/08993408.2022.2145549

2021 — **Roy, A.**, & Rambo-Hernandez, K. E. (2021). There's so much to do and not enough time to do it! A case for sentiment analysis to derive meaning from open text using student reflections of engineering activities. *American Journal of Evaluation, 42*(4), 559–576. https://doi.org/10.1177/1098214020962576

2020 — Kale, U., **Roy, A.**, & Yuan, J. (2020). To design or to integrate? Instructional design versus technology integration in developing learning interventions. *Educational Technology Research and Development, 68*(2473–2504). https://doi.org/10.1007/s11423-020-09771-8

2018 — Rambo-Hernandez, K. E., **Roy, A.**, Morris, M. L., Hensel, R. A. M., Schwartz, J. C., Atadero, R. A., & Paguyo, C. (2018, April). Using interactive theater to promote inclusive behaviors in teams for first-year engineering students: A sustainable approach. Paper presented at the 2018 CoNECD - The Collaborative Network for Engineering and Computing Diversity Conference, Crystal City, Virginia. https://doi.org/10.18260/1-2–29592

Richards-Babb, M., Curtis, R., Ratcliff, B., **Roy, A.**, & Mikalik, T. (2018). General chemistry student attitudes and success with use of online homework: Traditional-responsive versus adaptive-responsive. *Journal of Chemical Education, 95*(5), 691–699. https://doi.org/10.1021/acs.jchemed.7b00829

Miller, T. A., Carver, J. S., & **Roy, A.** (2018). To go virtual or not to go virtual, that is the question: A comparative study of face-to-face versus virtual laboratories in a physical science course. *Journal of College Science Teaching, 48*(2), 59–67. https://www.jstor.org/stable/26616271

2014 Davis, J. D., Smith, D. O., **Roy, A. R.**, & Bilgic, Y. K. (2014). Reasoning-and-proving in algebra: The case of two reform-oriented U.S. textbooks. *International Journal of Educational Research, 64*, 92–106. doi:10.1016/j.ijer.2013.06.012

2012 **Roy, A. R.**, Hobson, K. A., & Coryn, C. L. S. (2012). What's in a Scriven number? *Journal of MultiDisciplinary Evaluation, 8*(19), 41–45. https://doi.org/10.56645/jmde.v8i19.372

BOOK CHAPTERS

2017 **Roy, A. R.** (2017). Social network analysis: Finding meaning in connections. In T. C. Ahern (Ed.), *Social media*. Hauppauge, NY: Nova Science.

GRANTS

2024 Mining for Meaning: Uniting Machine Learning and Qualitative Research to Simulate Raters in Text Analysis
National Science Foundation: NSF 19-575: Methodology, Measurement, and Statistics (MMS) - *Pending* *Bloomington, IN*

Roy, A.R. - Principal investigator

INVITED CONTRIBUTIONS

2012 What is a Scriven number?
The American Evaluation Association Newsletter

Roy, A.R., Hobson, K.A., & Coryn, C.L.S.

EVALUATIONS

ACTIVE

Current | 2023

Lead Program Evaluator & Methodologist - Healthy Start Initiative (*HRSA-19-049* Total Award: *$5,470,000*)

West Virginia University Research Corporation — Morgantown, WV

- Created and validated survey instruments to assess participant experiences, collecting data from more than 50 individuals.
- Conducted evaluation studies of program activities and authored the evaluation section of the HRSA annual report, as well as a related federal funding renewal proposal.
- Designed and led program-level evaluations, including process, monitoring, and impact studies.
- Developed and implemented a mixed-methods study to analyze longitudinal and cross-sectional data, focusing on smoking cessation among mothers and the experiences of new or distant caregivers and fathers.

Current | 2022

Program Evaluator & Mixed Methods Analyst - Teaching Science with Computational Thinking: Preparing Preservice Elementary Educators of the Future STEM Workforce (*2019-NSF 2142274* Total Award: *$294,958.00*)

West Virginia University — Morgantown, WV

- Administered process evaluations for all program activities.
- Created and implemented a longitudinal mixed-method study to analyze more than 50 open-ended survey responses and six interview transcripts, using methods such as HDBSCAN/t-SNE, k-Means, and PCA alongside thematic and content analyses.
- Developed and validated survey instruments to track progress and evaluate the impact of program activities.
- Prepared detailed internal evaluation reports and developed summary documents for federal reporting.

COMPLETED

2022 | 2017

Community Program Evaluator & Data Scientist - WVCTSI: West Virginia Clinical and Translational Science Institute (*2017-NIH 2U54GM104942-02* Total Award: *$20,000,000*)

West Virginia University — Morgantown, WV

- Advised multiple aspiring graduate students in social data science as they developed and completed independent research projects.
- Analyzed datasets using using multiple guiding the direction and activities of eight distinct programs.
- Authored quarterly reports as well as internal and external annual evaluation documents, disseminated both in print and through interactive formats developed in Rmarkdown.
- Created data visualizations and developed Shiny applications to support internal and public data exploration, including social network analysis for research collaborations, integration of NCBI API data with WVCTSI grant records, and dissemination of findings and practice changes within and beyond West Virginia.
- Designed and disseminated tailored Qualtrics surveys reaching over 5,000 individuals, with user experience and visual design enhanced through HTML, CSS, and JavaScript.
- Led local and multi-site evaluation studies that informed policies across five core medical research and health outreach efforts.

2020 | 2021

Program Evaluator - Appalachian Gerontology Experiences - Advancing Diversity in Aging Research (*2020-NIH 1R25AG059558-01A1* Total Award: *$678,000*)

West Virginia University — Morgantown, WV

- Developed and distributed two customized interactive surveys using Qualtrics, with user experience and visual design enhanced through HTML, CSS, and JavaScript, to collect responses on programmatic activities from a targeted group of 24 students.
- Led three distinct evaluative studies focusing on student efficacy, engagement, and motivation
- Produced an external evaluation summary for federal reporting.

2020 | 2022

Research Methods Advisor & Specialist - Research Initiative: A Holistic Cross-Disciplinary Project Experience as a Platform to Advance the Professional Formation of Engineers (*2019-NSF 1927232* Total Award: *$200,000*)

West Virginia University — Morgantown, WV

- Conducted longitudinal studies on the experiences of 30 undergraduate students in specialized interdisciplinary courses that integrated social science and engineering, using both surveys and focus group discussions.
- Mentored engineering faculty members and graduate students in the implementation of research methodologies.

2018 | 2017

Program Evaluator - Stepping UP with Avenue: Progress Monitoring: A Software Suite Helping Teachers Improve Literacy Progress For Deaf/Hard Of Hearing Students (*2017-ED H327S170012* Total Award: *$2,470,440*)
Pennsylvania State University — State College, PA

- Evaluated digital tools developed for assessing and engaging individuals who are deaf or hard of hearing.
- Led a multi-site evaluation that examined program delivery, participant engagement, and outcomes.
- Produced a detailed external evaluation brief for federal reporting.

2018 | 2017

Lead Program Evaluator - Cultivating Inclusive Identities of Engineers and Computer Scientists: Expanding Efforts to Infuse Inclusive Excellence in Undergraduate Curricula (*2017-NSF 1725880* Total Award: *$2,000,000*)
West Virginia University — Morgantown, WV

- Applied longitudinal NLP text mining techniques such as concordance, LDA topic modeling, and sentiment analysis to analyze and summarize feedback from over 3,000 first-year engineering students concerning grant-related class activities.
- Carried out longitudinal surveys with over 50 faculty participants to understand expectations, gather input for improvement, and track changes in DEI attitudes and perceptions.
- Created a wide range of static and interactive data visualizations for stakeholder exploration, internal and external reporting, and use in presentations and publications.
- Contributed to multiple journal articles and academic conference presentations.
- Performed personnel evaluations of principal investigators to support broader program assessment efforts.
- Prepared internal evaluation reports and authored the evaluation section of the annual report submitted to the NSF.

2020 | 2017

Program Evaluator - GAUSSI: Generating, Analyzing, and Understanding Sensory and Sequencing Information: A Trans-Disciplinary Graduate Training Program in Biosensing and Computational Biology (*2017-NSF 1450032* Total Award: *$3,013,779*)
Colorado State University — Fort Collins, CO

- Carried out 32 interviews and focus groups, both cross-sectional and longitudinal, using unstructured and semi-structured formats to explore the experiences of students and faculty.
- Conducted process evaluations for multiple grant-funded programs, resulting in improved member tracking, greater program efficiency, and increased participant satisfaction.
- Created and validated instruments to assess students' ability to communicate research findings to a general audience.
- Designed semi-annual adaptive and interactive Qualtrics surveys enhanced with HTML/CSS/JavaScript, securing feedback from over 100 students and faculty with a 95% response rate.
- Generated data visualizations for assessment and longitudinal studies, bolstering inferential statistical analyses to identify trends and support programmatic enhancements, retention strategies, and satisfaction initiatives.
- Produced internal evaluation reports and prepared summaries for use in external communications and federal reporting.

PRESENTATIONS

2022

Let's get sentimental: Machine learning aided data analysis for large qualitative data sets
American Evaluation Association Annual Conference — New Orleans, LA

Seidel, T., Ferguson, C.F., & **Roy, A.R.**

2022

Best of both worlds: Affordances of mixing machine learning and qualitative content analysis
American Educational Research Association Annual Meeting — San Diego, CA

Roy, A.R., Ferguson, C.F., Curtis, R., & Babb-Richards, M.

2020

These aren't random words just strung together?: Using machine learning and pretty visualizations to discover topics in articles.
American Evaluation Association Annual Conference — *virtual*

Roy, A.R.

2019 Little fish in a big pond, only fish in a little pond: How roles shape our identities as evaluators.
American Evaluation Association Annual Conference Minneapolis, MN

Loomis, D.L., Mikalik, T.L., Curtis, R., **Roy, A.R.**, & Bernstein, M.

2019 Evolving program logic models to meet shifting program needs: The case of WV Clinical Translational Science Institute.
American Evaluation Association Annual Conference Minneapolis, MN

Curtis, R., **Roy, A.R.**, Bernstein, M, Loomis, D.L., & Mikalik, T.L.

2019 The value of external evaluators when building clinical translational research infrastructure.
American Evaluation Association Annual Conference Minneapolis, MN

Curtis, R., **Roy, A.R.**, Bernstein, M, Loomis, D.L., & Mikalik, T.L.

2019 Using associated networks to evaluate content within courses.
American Evaluation Association Annual Conference Minneapolis, MN

Roy, A.R., Kale, U, & Yuan, J.

2019 Why is it that writers write but fingers don't fing? Using machine learning and lexemes to make sense of nonsense.
American Evaluation Association Annual Conference Minneapolis, MN

Roy, A.R., Curtis, R., Mikalik, T.L., Loomis, D.L., & Bernstein, M.

2019 Iscovering the underlying meaning behind *get me off your f***ing mailing list?* and most other narratives.
American Evaluation Association Annual Conference Minneapolis, MN

Roy, A.R., Curtis, R., Mikalik, T.L., Loomis, D.L., & Bernstein, M.

2019 Assessing for improvement: The use of artificial intelligence to uncover potential differential impact of assignments.
American Evaluation Association Annual Conference Toronto, CN

Roy, A.R. & Rambo-Hernandez, K.

2018 That's a pretty picture of dots and lines but what does it mean?: A Q&A session with the Social Network Analysis TIG leaders.
American Evaluation Association Annual Conference Cleveland, OH

Roy, A.R., Durland, M.M., Woodland, R., & Phillips, G.

2018 Navigating buy-in and shifting evaluation needs over time in NIG Clinical Translational Research Award.
American Evaluation Association Annual Conference Cleveland, OH

Curtis, R., **Roy, A.R.**, & Mikalik, T.L.

2018 Using a mixed methods evaluation to discover how an interactive theater based model stimulates inclusive behaviors in engineering.
American Evaluation Association Annual Conference Cleveland, OH

Roy, A.R., Rambo-Hernandez, K., Hensel, R.A., & Morris, M.L.

2018 Collaboration evaluation: Using social network analysis to reveal an active undiscovered network.
American Evaluation Association Annual Conference Cleveland, OH

Roy, A.R., Curtis, R., & Mikalik, T.L.

2018 Examining the past and looking forward: The future of evaluation theory and use.
American Evaluation Association Annual Conference Cleveland, OH

Roy, A.R. & Hobson, K.A.

2017 Transforming graduate STEM Education: A theory-driven evaluation of the GAUSSI National Science Foundation Research Training (NRT) Program.
American Evaluation Association Annual Conference Washington, DC

Roy, A.R., Hernandez, P.A., Chen, T., & Paguyo, C.

2017 Program evaluation for everyone! - Constructing an online foundational course for capacity building using theorists as a focus.
American Evaluation Association Annual Conference Washington, DC

Roy, A.R. & Curtis, R.P.

2017 Three stages down! Exploring the criteria for the next generation of evaluation theorists through social network analysis.
Hawaii-Pacific Evaluation Association Annual Conference Kane'ohe, HI

Roy, A.R. & Hobson, K.A.

2017 Content in the background: Using evaluation theorists as the principal motivator for foundational evaluation courses.
Hawaii-Pacific Evaluation Association Annual Conference Kane'ohe, HI

Roy, A.R. & Curtis, R.P.

2015 Survey says! Students getting tired of surveys.
National Academic Advising Association Annual Conference Las Vegas, NV

Roy, A.R. & Goetz, H.L.

2013 Influences of Hierarchical Linear Modeling in evaluation.
Aotearoa New Zealand Evaluation Association Annual Conference Auckland, NZ

Hobson, K.A., **Roy, A.R.** & Coryn, C.L.S.

2012 Survey sample methods: Evaluators' toolbox refreshment.
American Evaluation Association Annual Conference Minneapolis, MN

Hobson, K.A., **Roy, A.R.** & Coryn, C.L.S.

TEACHING EXPERIENCE

EVALUATION, MEASUREMENT, AND RESEARCH METHODS (2016 - 2023)

2020 | 2018 *Data Visualization*
West Virginia University Morgantown, WV

2020, 2018

2017 *Educational Psychology*
West Virginia University Morgantown, WV

2016 — *Educational Research*
West Virginia University — Morgantown, WV

2022 | 2016 — *Introduction to Research*
West Virginia University — Morgantown, WV
2022, 2018, 2017, 2016

2020 | 2018 — *Measurement/Evaluation in Educational Psychology*
West Virginia University — Morgantown, WV
2022, 2020, 2018

2019 | 2017 — *Mixing Research Methodologies*
West Virginia University — Morgantown, WV
2022, 2019, 2018, 2017

2023 | 2017 — *Program Evaluation*
West Virginia University — Morgantown, WV
2023, 2022, 2021, 2020, 2019, 2018, 2017

2021 | 2017 — *Social Network Analysis*
West Virginia University — Morgantown, WV
2021, 2017

2021 | 2017 — *Statistical Methods 1*
West Virginia University — Morgantown, WV
2021, 2020, 2019, 2018, 2017

2022 | 2020 — *Survey Research*
West Virginia University — Morgantown, WV
2022, 2020

∞ MATHEMATICS (2005 - 2015)

2008 — *Business Calculus*
Central Michigan University — Mount Pleasant, MI

2009 — *College Algebra*
Central Michigan University — Mount Pleasant, MI

2012 — *Discrete Mathematics*
Central Michigan University — Pittsburgh, KS

2014 — *Elementary Statistics*
Central Michigan University — Pittsburgh, KS

2010 | 2009 — *Foundations of Statistics*
Central Michigan University — Pittsburgh, KS
2009, 2010

2008 | 2007 — *Intermediate Algebra*
Central Michigan University — Mount Pleasant, MI
2007, 2008

2007 — *Integral Calculus*
Michigan Technological University — Houghton, MI

2015 | 2014 — *Linear Algebra*
University of Kansas — Lawrence, KS
2014, 2015

2013 — *Mathematical Thinking Grades 6-12*
Western Michigan University — Kalamazoo, MI

2014 | 2013 — *Mathematics Curriculum Grades 6-12*
Western Michigan University — Kalamazoo, MI
2013, 2014

2005 — *Multivariable Calculus*
Michigan Technological University — Houghton, MI

2007 | 2006 — *Single Variable Calculus*
Michigan Technological University — Houghton, MI
2006, 2007

SERVICE

2022 | 2013 — Associate Editor
Journal of MultiDisciplinary Evaluation

MEMBERSHIPS

2022 | 2012 — American Evaluation Association